

Running AIs on Your Own Device



CU Boulder AI Club Workshop Series

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Outlook

Goals:

1. Learn why running AI on your laptop can be helpful and safer
2. Learn how to download and use your own AI models
3. Project work time!

Why Run AI Models On-Device?

- **Privacy:** the prompts and data that you upload never go a company or outside of your own computer
- **Speed:** prompts don't need to be uploaded to the internet – very fast!
- **Control:** you are able to tune the models to behave exactly as you intend
 - System prompt
 - Temperature
 - Threads
 - Weights
- **Offline and Free** – no usage limits and you don't need wifi!

What's different compared to online?

- Models like GPT 5 have somewhere between 2-10 **trillion** parameters ... But our laptops can handle maybe 5 billion params. (0.1%)
- We will have to use smaller models! Great open-source models from Meta:
 - Tiny-Llama-1.1B
 - Meta-Llama-3.2-3B-Instruct

(LLaMA: Large Language Model Meta AI)

(Open source – completely free and public)



How can small models still perform?

Quantization: lowering the “resolution” of a model – similar to an image

- AI models are made of tons of vectors and matrices – huge blocks of decimal numbers
- These decimals usually have 32 decimal places, but we can discard some
- Q2 = 2 decimal places, Q4 = 4 decimal places, etc...

We don't count
non-decimals (3)

3.14159265358979323846264338327950

Q4

Q8

Q2

Actually running the models...

- We will use llama.cpp – another great open source program
1. Navigate to this link: tinyurl.com/yc4k9yj8
 2. Download llama.cpp:

tes ago
ub-actions
27
19cf 
are ▾

b7027

Latest

disable rms norm mul rope for chips with no fp16 rte (#17134)

▼ **Assets** 16

 [cudart-llama-bin-win-cuda-12.4-x64.zip](#)

 [llama-b7027-bin-macos-arm64.zip](#)

 [llama-b7027-bin-macos-x64.zip](#)

 [llama-b7027-bin-ubuntu-s390x.zip](#)

Windows click here!

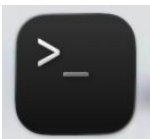
Mac click here!

Mac (2020 or older) click here!

Linux click here!

Installing, continued...

Mac:



1. Open the “terminal” app
2. Type this in to download the model (Tiny-Llama):

```
llama-server -hf  
TheBloke/TinyLlama-1.1B-Chat-v1.0-GGUF  
--model  
tinyllama-1.1b-chat-v1.0.Q4_K_M.gguf
```

Wait for it to download...

Open <http://127.0.0.1:8080> in your browser and enjoy!

Windows:



1. Find your llama.cpp folder in File Explorer.
2. Right-click inside the folder (not on a file).
3. Select “Open in Terminal” or “Open PowerShell window here.”
4. Type this in to download the model:

```
.\llama-server.exe -hf  
TheBloke/TinyLlama-1.1B-Chat-v1.0-GGUF  
--model  
tinyllama-1.1b-chat-v1.0.Q4_K_M.gguf
```

Wait for it to download...

Open <http://127.0.0.1:8080> in your browser and enjoy!

Playing around

- Can also try this command for a slightly **bigger model (Llama-3.2b-Instruct)**:

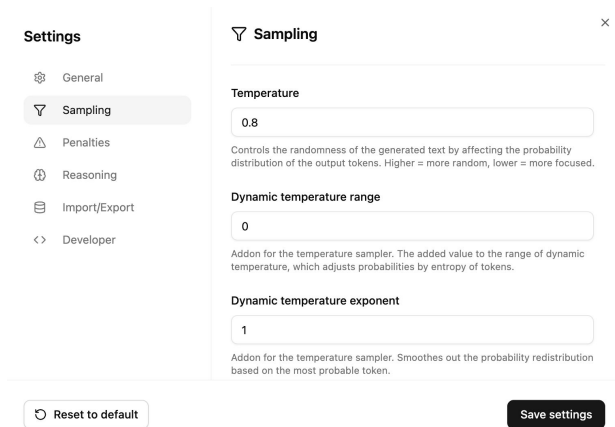
Mac:

```
llama-server -hf bartowski/Llama-3.2-2B-Instruct-GGUF --model  
llama-3.2-2b-instruct.Q4_K_M.gguf
```

Windows:

```
.\llama-server.exe -hf bartowski/Llama-3.2-2B-Instruct-GGUF --model  
llama-3.2-2b-instruct.Q4_K_M.gguf
```

- Also try different settings!
 - Temperature (randomness)
 - System prompt
 - Penalties
 - Change weights (ask ChatGPT)



The screenshot shows the 'Settings' window with the 'Sampling' tab selected. The 'Temperature' is set to 0.8, with a description: 'Controls the randomness of the generated text by affecting the probability distribution of the output tokens. Higher = more random, lower = more focused.' The 'Dynamic temperature range' is set to 0, with a description: 'Addon for the temperature sampler. The added value to the range of dynamic temperature, which adjusts probabilities by entropy of tokens.' The 'Dynamic temperature exponent' is set to 1, with a description: 'Addon for the temperature sampler. Smooths out the probability redistribution based on the most probable token.' At the bottom, there are buttons for 'Reset to default' and 'Save settings'.

Settings

- General
- Sampling**
- Penalties
- Reasoning
- Import/Export
- Developer

Sampling

Temperature

0.8

Controls the randomness of the generated text by affecting the probability distribution of the output tokens. Higher = more random, lower = more focused.

Dynamic temperature range

0

Addon for the temperature sampler. The added value to the range of dynamic temperature, which adjusts probabilities by entropy of tokens.

Dynamic temperature exponent

1

Addon for the temperature sampler. Smooths out the probability redistribution based on the most probable token.

Reset to default Save settings

Thank you everyone!

Thank you for a great semester of workshops!

Next semester:

- Hackathons?
- More workshops?
- More projects?
- AI safety and ethics?